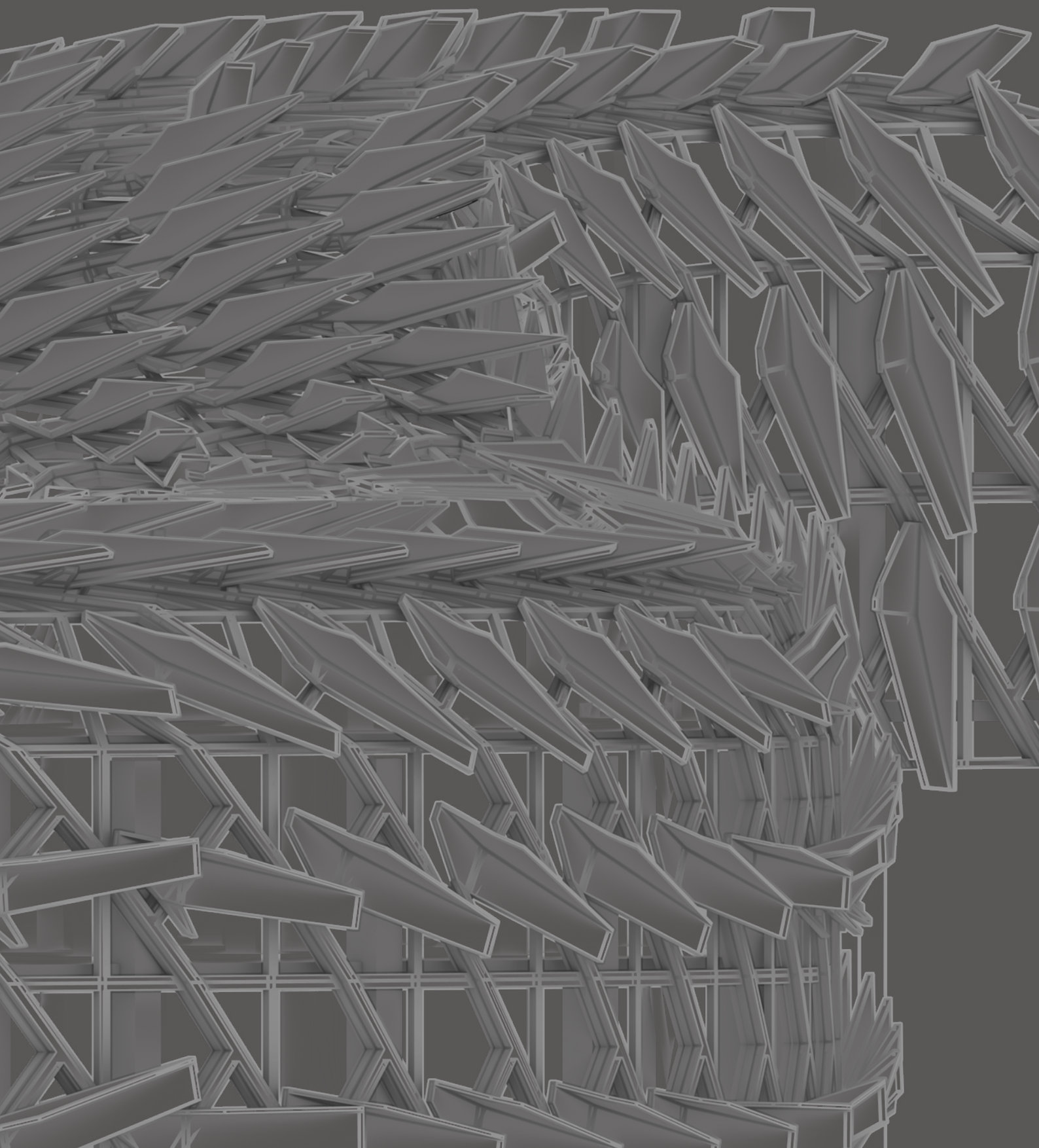
COMPUTATIONAL PORTFOLIO

Sara Elimam | Selected Works | 2025 - 2026





Sara Elimam

About Me

I am an Architectural Engineer specializing in computational design, building envelope optimization, and AI integration. Holding a Master's from Politecnico di Milano, my objective is to embed rigorous environmental simulation into standard architectural pipelines, seamlessly aligning conceptual morphology with measurable, data-driven performance.

My work bridges the gap between advanced predictive modeling and practical technical execution. Recently, I developed an image-based machine learning framework (utilizing DEMs and CNNs) that bypasses time-intensive simulations to provide near real-time environmental feedback directly within Grasshopper. I apply this same focus on efficiency and precision to professional practice—including international projects like the Al Salam Bank Headquarters—where I integrate advanced parametric workflows with BIM platforms (Revit, ArchiCAD) to translate highly optimized, complex geometries into constructible realities.

Sara Elimam

Junior Computational Designer

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saraelimam.wordpress.com

Education

2023 - 2026

Building and Architectural Engineering

Laurea Magistrale at Politechnico Di Milano Italy

108/110 | Focus: Applied Machine Learning & Performance-Driven Design
Researched and developed an early-stage design tool bridging parametric modeling and machine learning. The workflow generated 1,000+ multi-objective façade simulations in Grasshopper and analyzed the design space using Pareto frontiers. These 3D morphologies were then encoded into Digital Elevation Models (DEMs) to train a Convolutional Neural Network (CNN). Achieving a mean R^2 of 0.955, the final tool accurately predicts environmental metrics, offering real-time feedback without heavy simulation engines.

2015 - 2021

Architecture

BSc Honors at University Of Khartoum Sudan

Honors | Second Class - Upper
Designed acoustic solutions for an in-campus lecture hall.
Conducted applied science calculations for room heat gain and A/C design.

Work Experience

2022 - 2023

Design Engineer

Mo.Siraj Engineering Co. Ltd Sudan

2022

Teaching Assistant

University of Khartoum Sudan

2018

Architectural Intern

Albarajoub Engineering Co. Ltd Sudan

Expertise

Algorithmic Process Optimization
Developing and using computational design tools to streamline design processes.

Multidisciplinary Design Integration
Collaborating with architects, engineers, and construction professionals to integrate computational design into the design and construction process.

Data-Driven Decision Support
Creating and managing digital models, simulations, and visualisations to support decision-making throughout the design and construction process.

Custom Workflow Automation
Developing custom tools and workflows to automate routine tasks and improve design quality and efficiency.

Continuous Technology Research
Staying up-to-date with the latest developments in computational design and related technologies.

Technical Stakeholder Communication
Communicating design concepts and solutions to clients and other stakeholders.

Skills

Python
Rhino 3D
Grasshopper
Revit Architecture
Rhin.Inside.Revit
AutoCAD
Kangaroo Physics
Ladybug Tools
Photoshop
InDesign
Illustrator
ArchiCAD
Lumion



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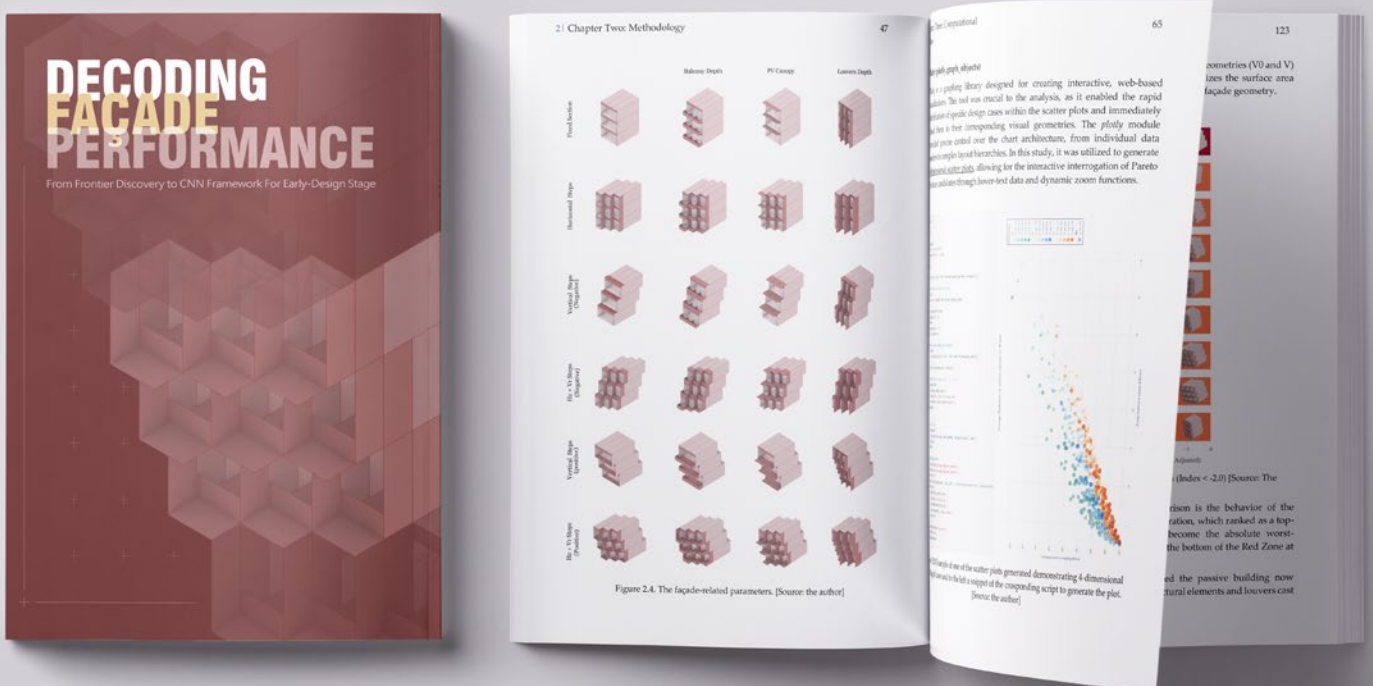
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Facade Lab

Laurea Magistrale Thesis

To check the full project Report on Polimi repository. Scan the QR code

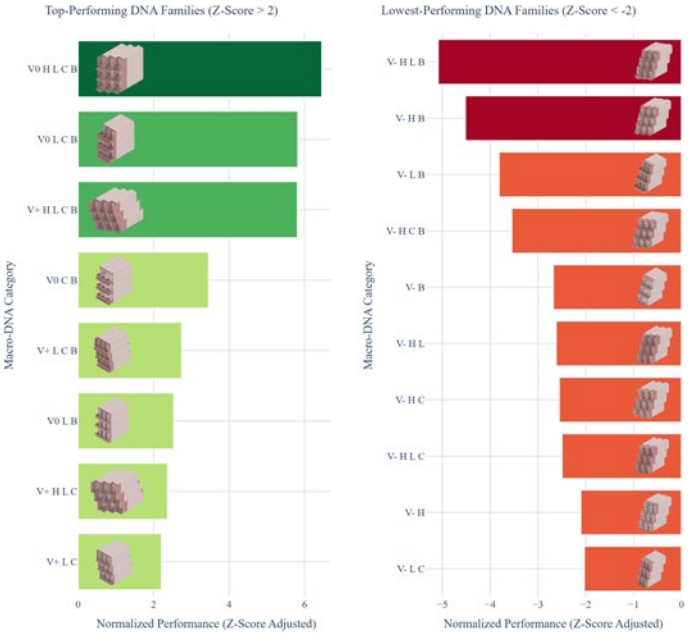


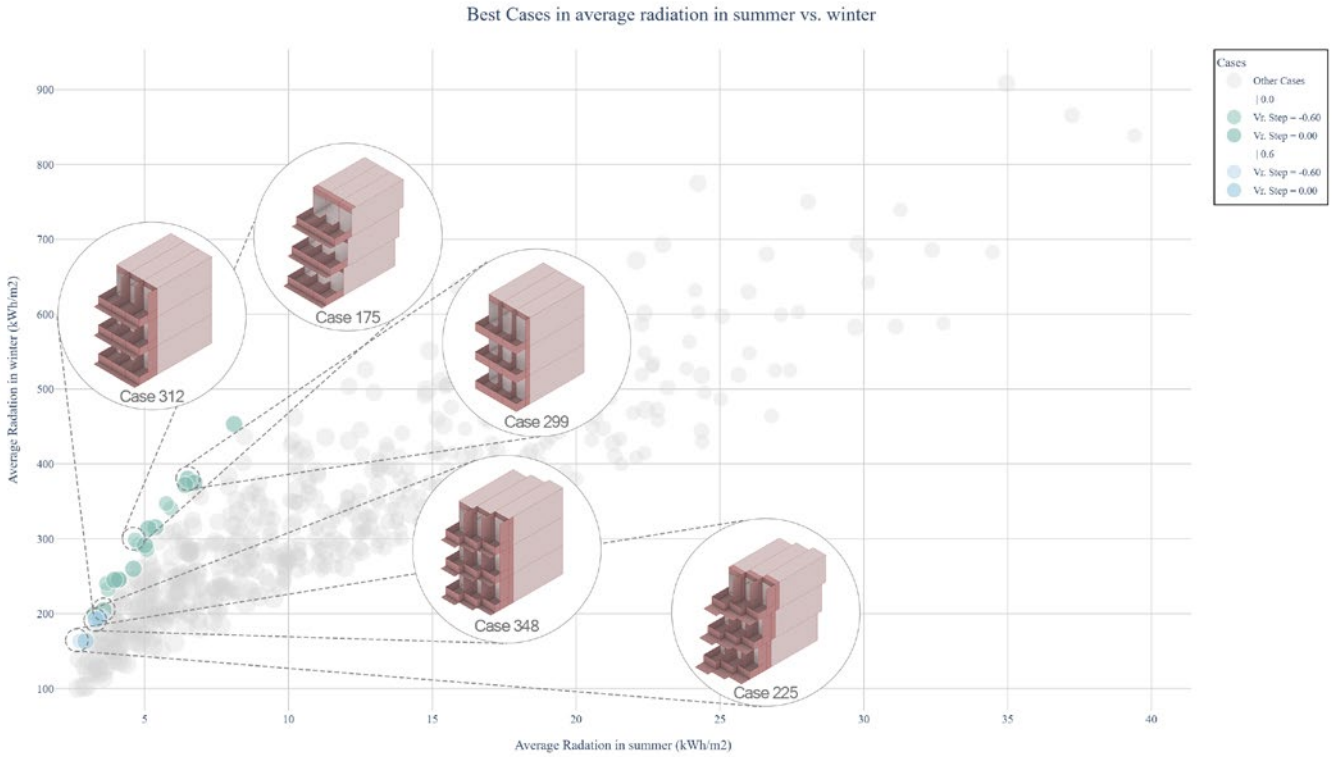
Thesis Collaboration with Hagar Hammad

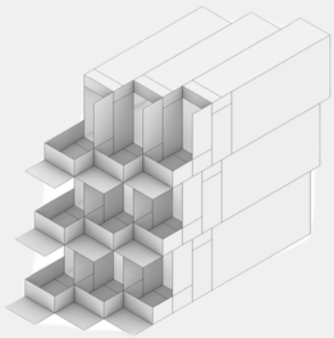
This thesis transforms façade design from simulation-heavy evaluation into AI-assisted exploration. Over 1,000 morphological variations were simulated to build a performance dataset, then encoded as Digital Elevation Models (DEMs) for a Convolutional Neural Network, achieving R^2=0.955 in predicting thermal, daylight, and active solar performance. The resulting framework enables near real-time environmental feedback, bringing data-driven sustainability directly into early-stage design.

Data Production and analysis

Data points were parametrically generated and analyzed through Pareto frontiers in distinct domains, each featuring contradicting environmental targets. The data was then grouped into specific categories based on architectural morphology. Finally, group Z-scores were applied to rank the data points' overall performance stability across these objectives. The front elevation of each generated case was then used to train the CNN Machine Learning model.

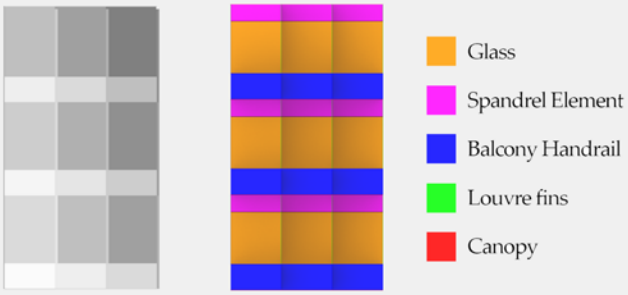






Back-end Development

Predictive accuracy R^2 = 0.95



1

Inputs and compression for compatibility

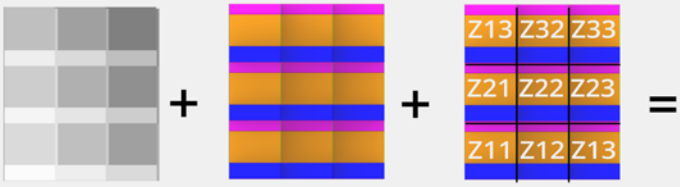
The system processes two types of inputs simultaneously, formatted to a standardized square images. The Digital Elevation Model (DEM) and the semantic color-coded map divided into five separate layers, each representing a distinct architectural feature.

Winter Radation
Summer Radation
sDA
ASE
Active Generation

Y = Normalised Performance vector

Dataset split:
70% Training
15% Validation
15% Test
Epochs used is 50
Batch size of 16
Best model is selected using validation loss

2

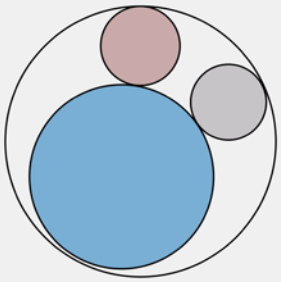


DEM Encoder | Multichannel Encoder | Zone Pooling

Geometric feature vector

Feature Fusion Head

4



3

Encoder, Fusion, and Regression head

The pipeline maps visual inputs (X)—combining geometric (X_dem) and spatial-semantic (X_sem) features—to a 5-performance indicator vector (Y). During training, the targets (Y) are normalized (Y_norm) to balance scales. A regression head processes the fused X representation to predict normalized outputs (Y_pred), which are denormalized during testing to recover physical units. The final model achieved high accuracy, yielding an average R^2 of 0.955

Regression Head

5

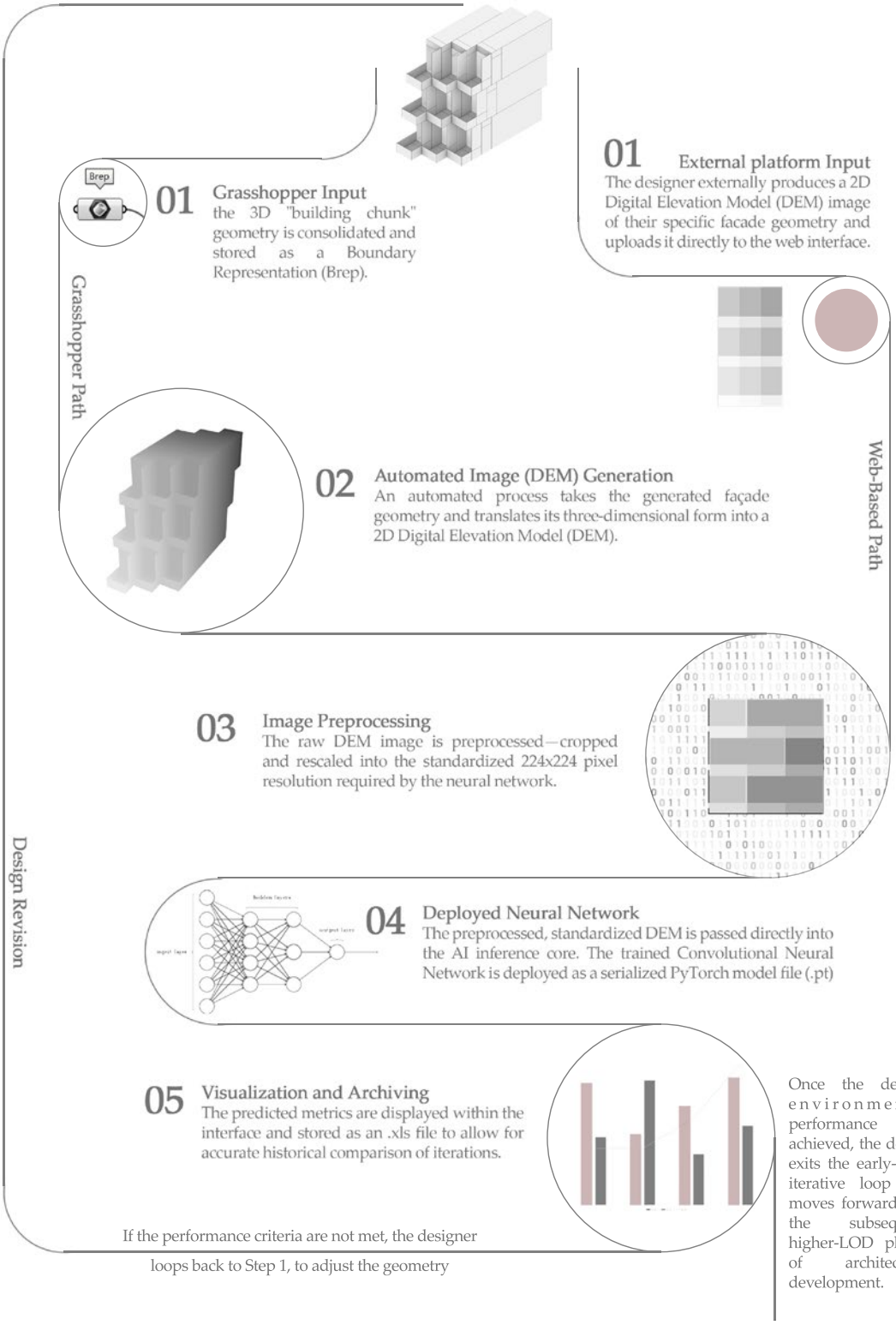
Predections

R^2 — the closer to one the better

MAE — the closer to zero the better

6

Front-end Development

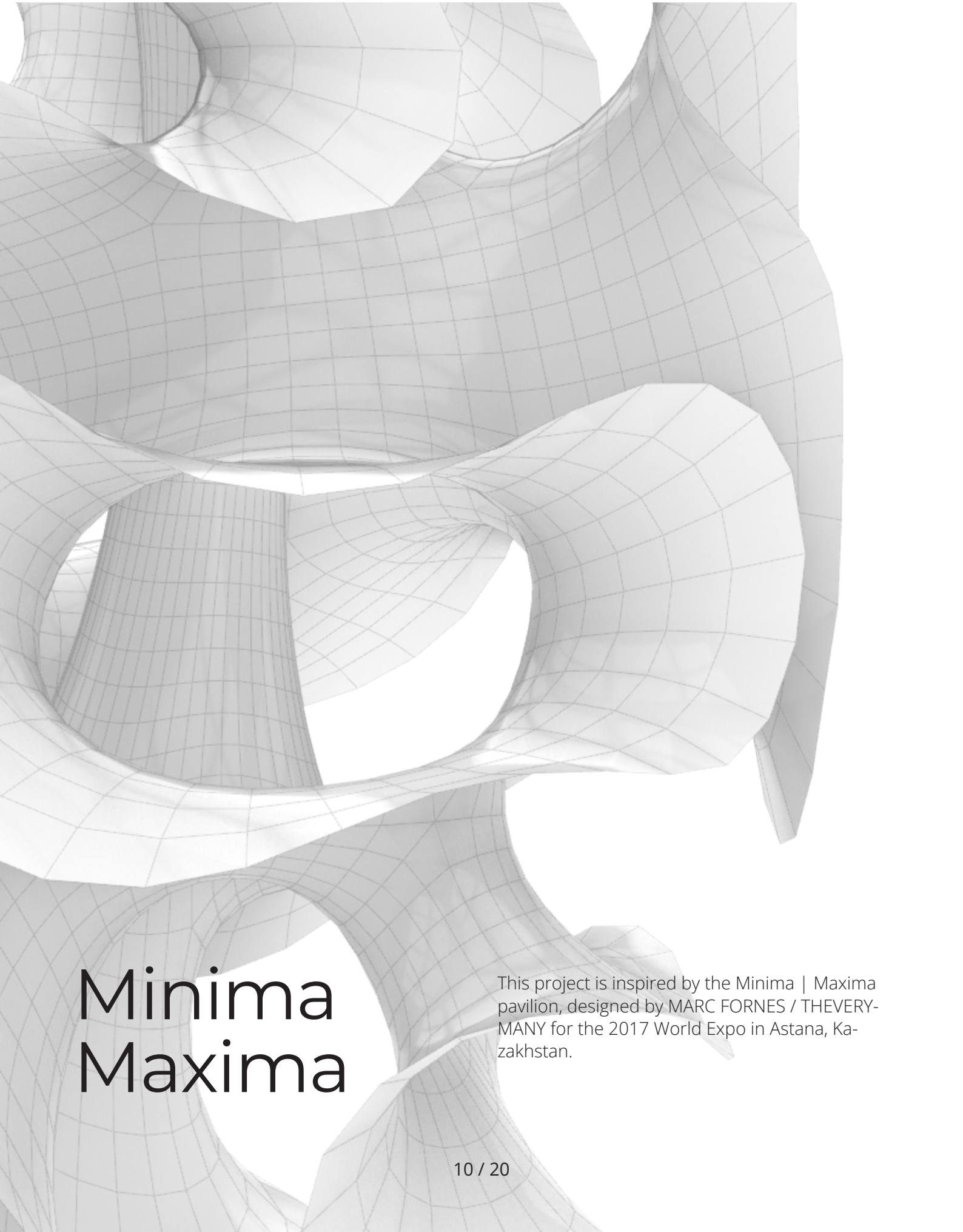


Click or scan the qr code to access the prototype! (Restart the space if error time encountered)

The framework deploys via two interfaces:

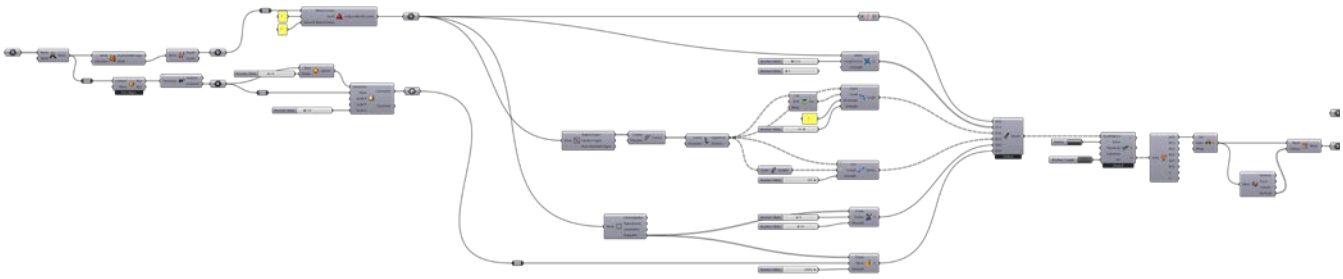
Web Dashboard:
A prototype providing rapid, ML-driven environmental predictions. It translates DEM Images into predicted values

Grasshopper Integration:
A custom native tool that automatically extracts 3D geometry, returning real-time performance feedback directly onto the design canvas.



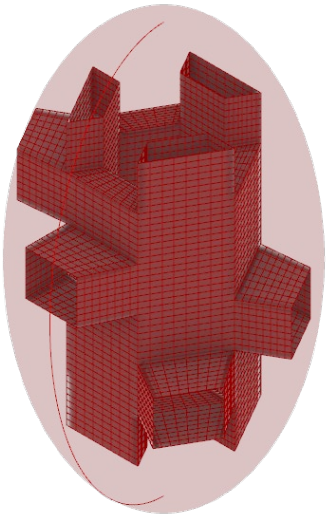
Minima Maxima

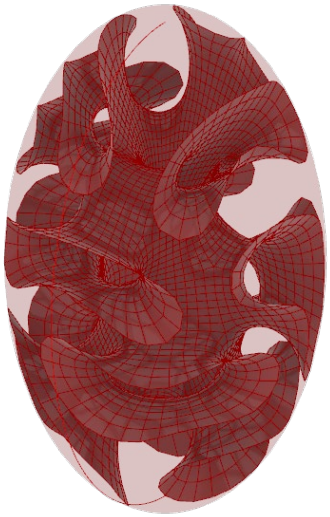
This project is inspired by the Minima | Maxima pavilion, designed by MARC FORNES / THEVERY-MANY for the 2017 World Expo in Astana, Kazakhstan.

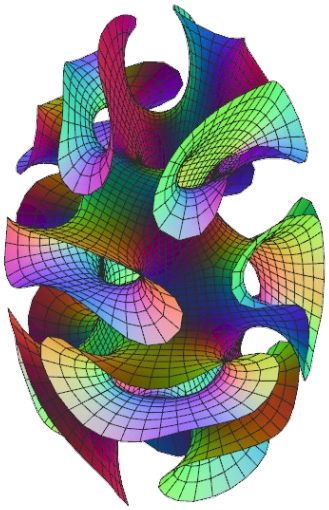


The initial form-finding phase utilizes Grasshopper and the Jellyfish plugin to generate a gyroid minimal surface. The geometry is selectively adjusted and then optimized through quad remeshing. By manually correcting the topology to ensure only even valences remain, the finalized base mesh can perfectly support continuous edge-to-edge structural strips.

ALGORITHMIC FORM-FINDING







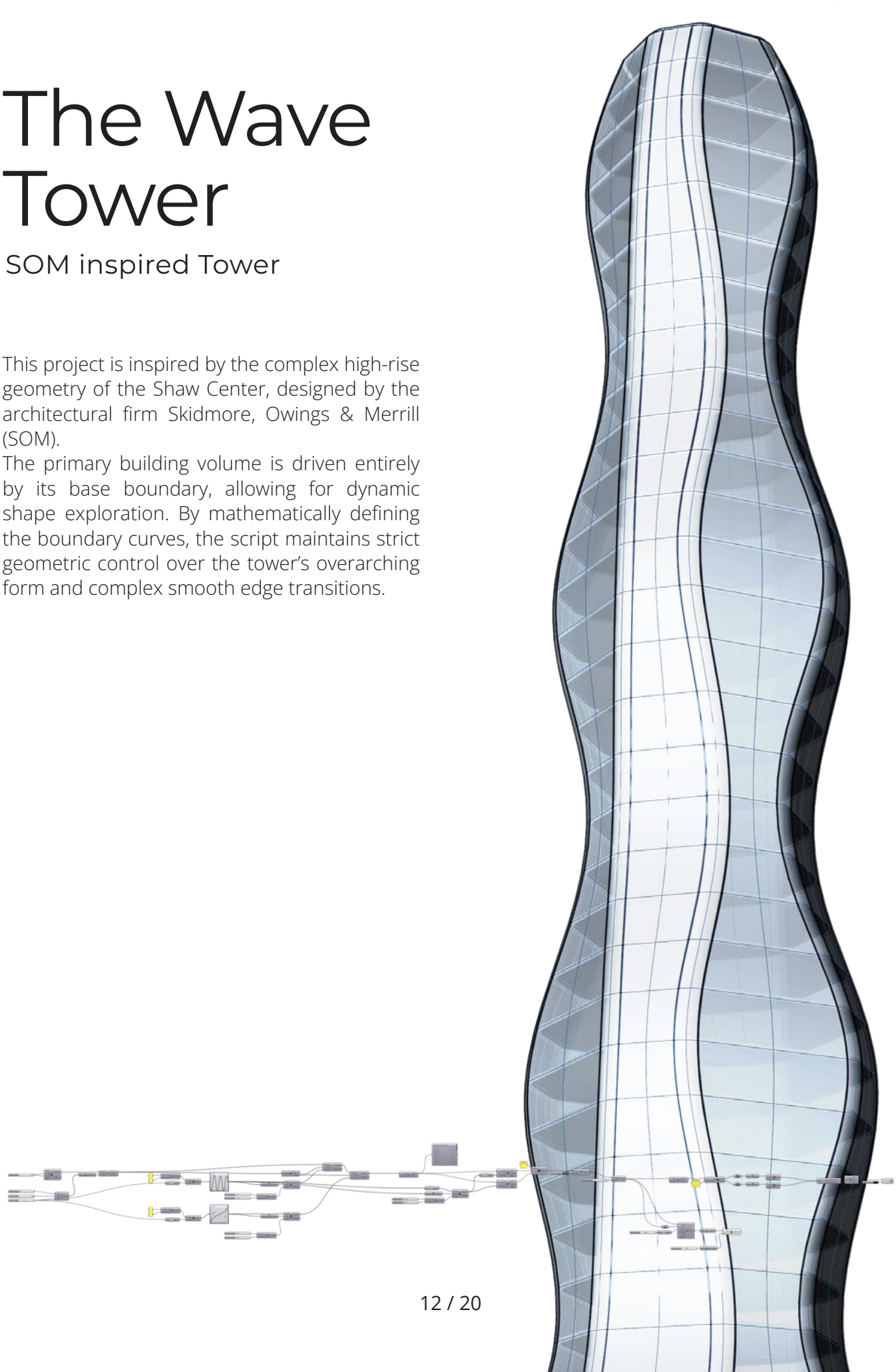
The second phase employs the Kangaroo solver to relax the structural mesh. Specialized goals prevent border collapse and ensure even face distribution. This physical simulation mitigates localized distortions, achieving optimal material continuity.

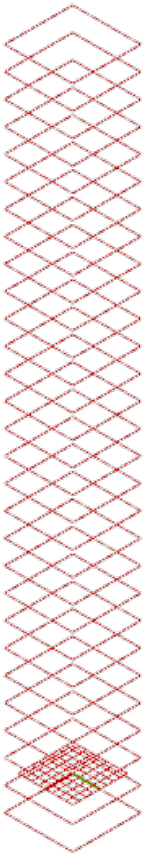
Finally, the geometry is rationalized into fabrication strips, partitioned into staggered connections for structural integrity. Flattened components are separated by precise tolerances and nested using OpenNest for ready-to-cut three-layer assemblies.

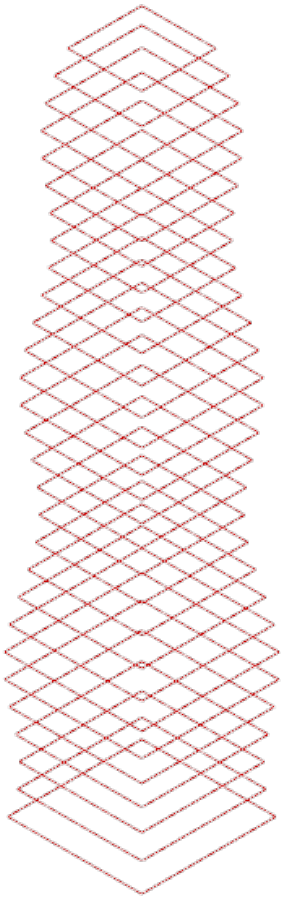
The Wave Tower

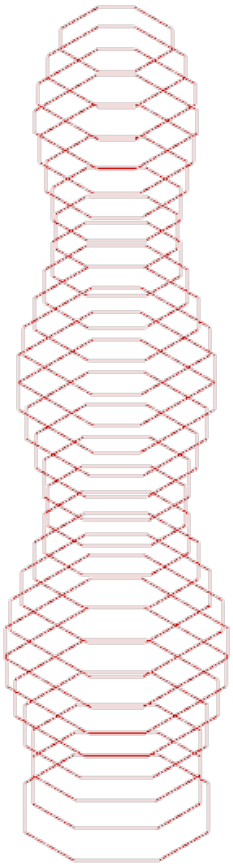
SOM inspired Tower

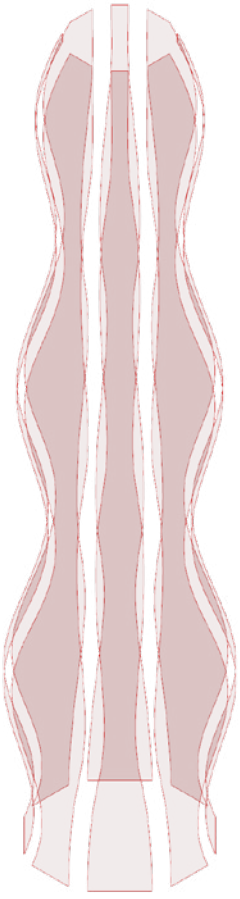
This project is inspired by the complex high-rise geometry of the Shaw Center, designed by the architectural firm Skidmore, Owings & Merrill (SOM).
The primary building volume is driven entirely by its base boundary, allowing for dynamic shape exploration. By mathematically defining the boundary curves, the script maintains strict geometric control over the tower's overarching form and complex smooth edge transitions.

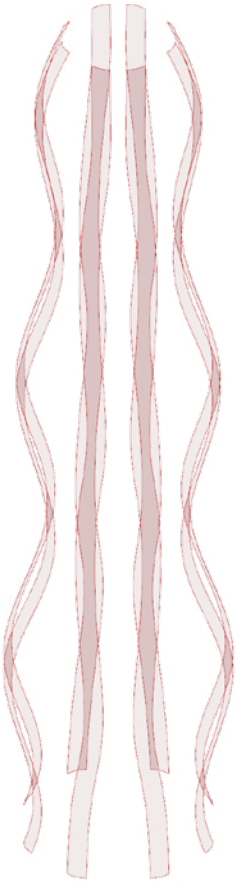












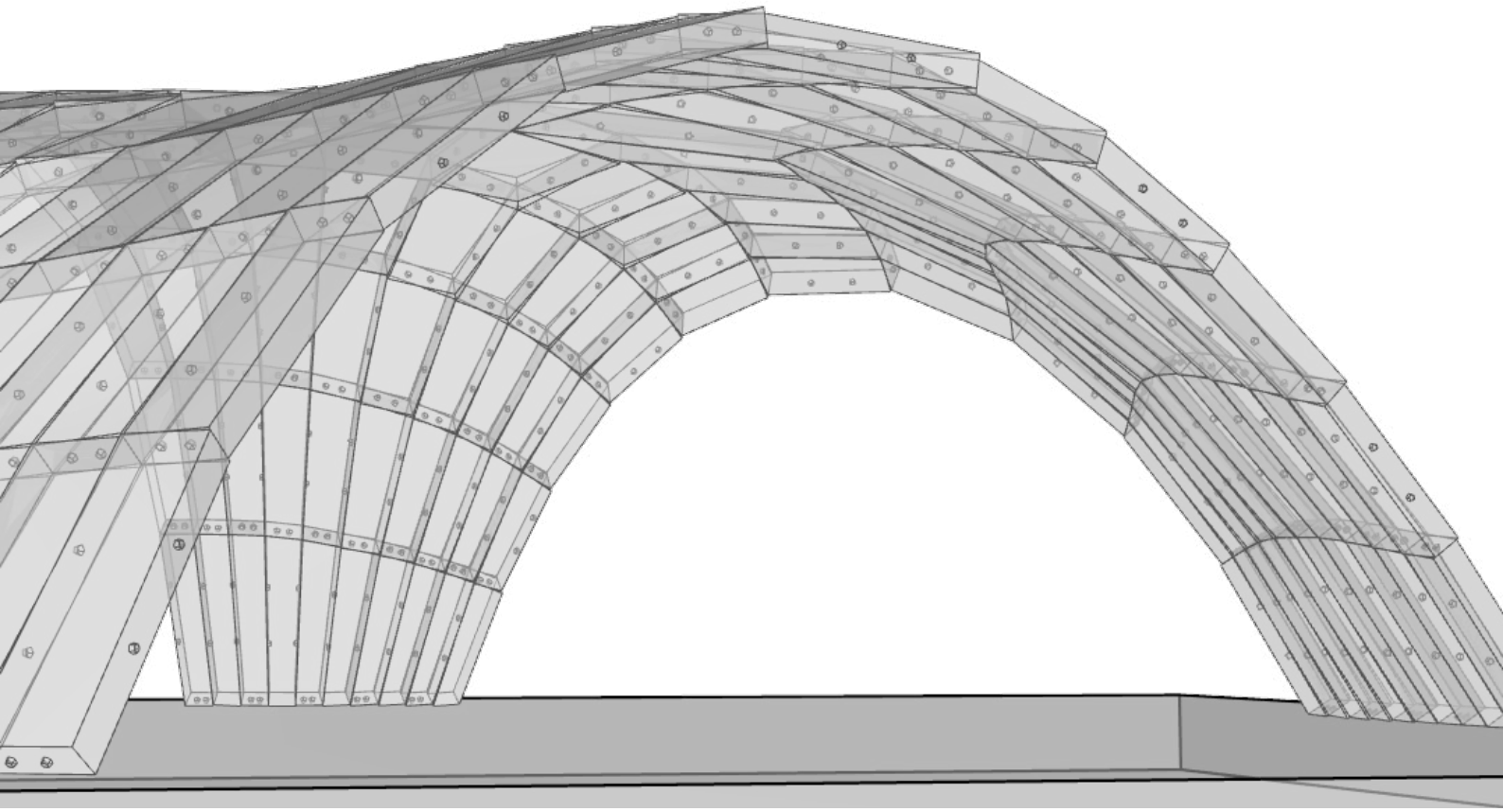
The Script Logic

The tower's massing is generated by defining a base polygonal boundary and filleting its vertices to create a continuous planar profile. This base curve is arrayed vertically along the Z-axis to establish the building's total elevation. To control the structural taper, a Grasshopper Graph Mapper evaluates the vertical domain, outputting specific scaling factors that non-uniformly resize each profile curve in the XY plane. A Loft operation then connects these parameterized sections, generating a single NURBS poly surface that acts as the primary envelope.

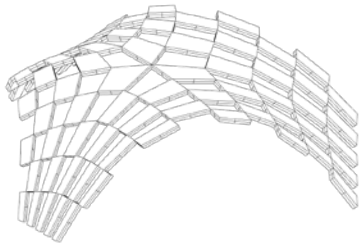
Check the Full report on my LinkedIn post!

Compression-only vault

This project demonstrates a complete computational workflow for designing a compression-only masonry vault. Bridging structural physics with manufacturing constraints, the script translates a 2D footprint into a spatial shell comprised entirely of planar, interlocking blocks ready for digital fabrication.

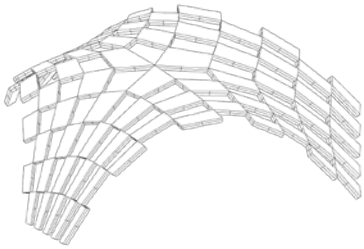


06



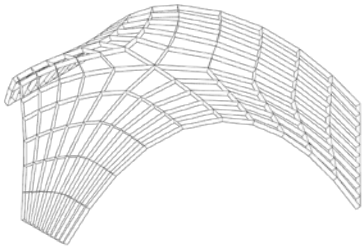
A checkerboard algorithm dispatches alternating blocks. Local UV normals generate extruded male pegs and boolean female cavities for dry assembly.

05



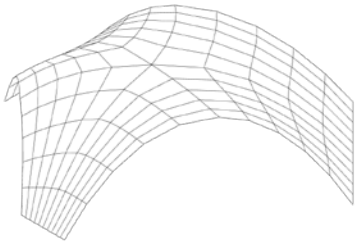
A secondary Kangaroo solver, paired with the Ngons plugin, acts on the data tree to force every individual quad-face flat within a 1mm tolerance.

04



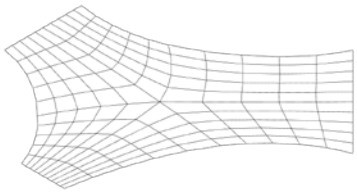
Standard curved surfaces are inefficient for flat-sheet milling. The primary shell is offset to establish uniform material thickness for the blocks.

03



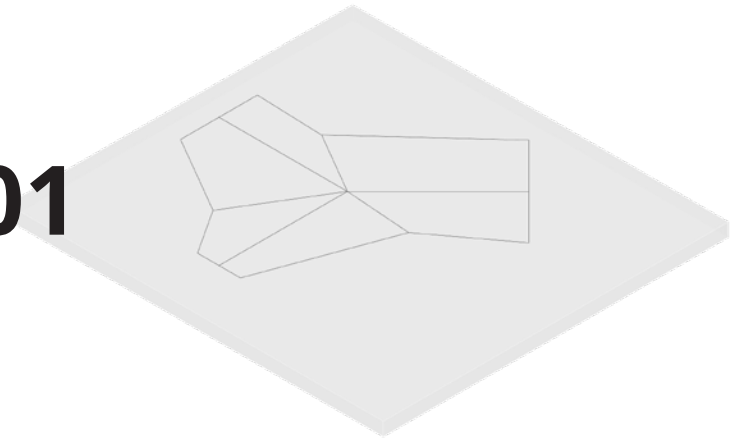
Simulated inverted gravity is applied to the anchored grid. This inflates the flat topology into a structurally optimized, compression-only 3D vault.

02



Using the Kangaroo physics solver, the flat grid undergoes dynamic relaxation, balancing internal tension and regularizing the mesh edge lengths.

01



A 2D skeletal footprint is modeled and populated with 4-point polylines. This ensures the base geometry subdivides cleanly into a strict quad-mesh.

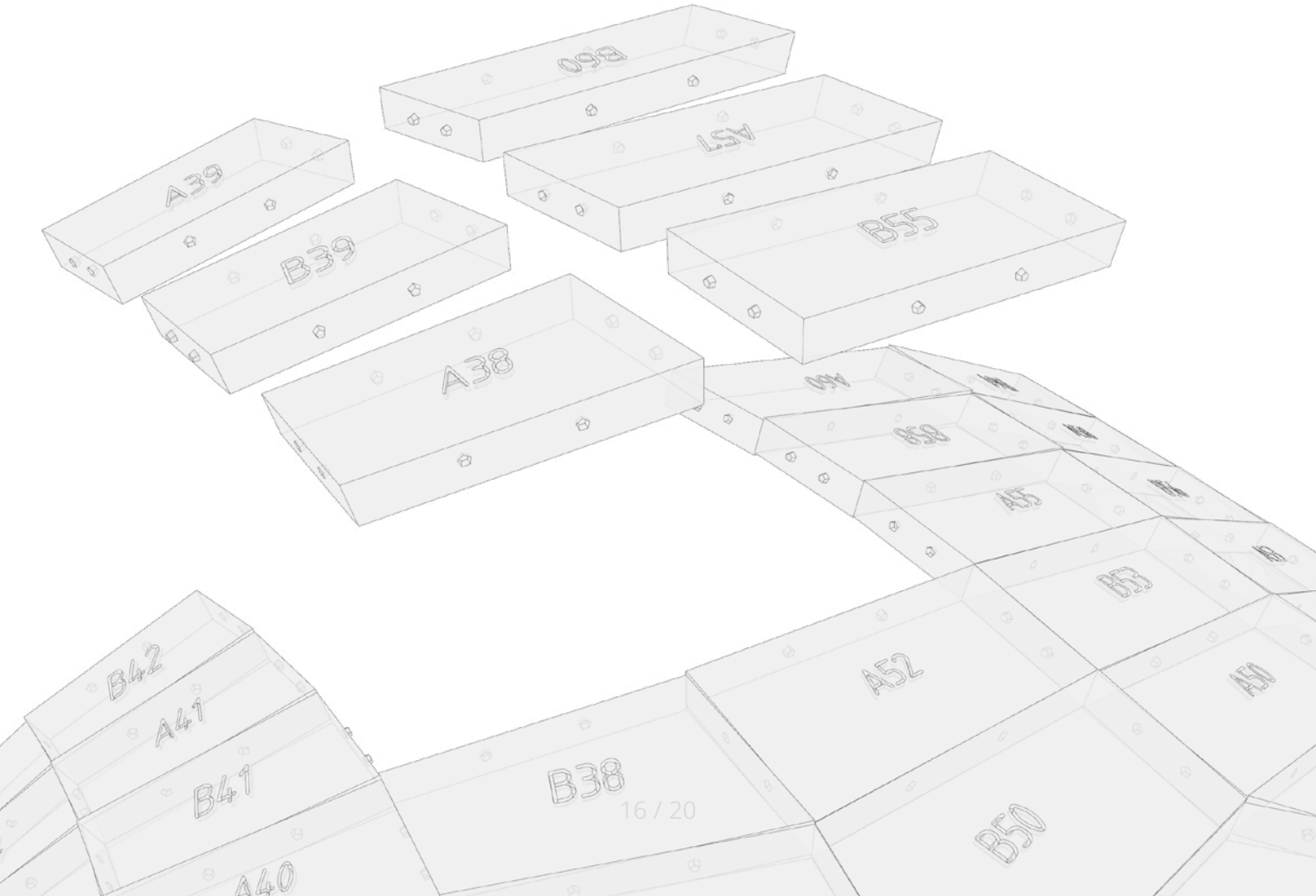
Precedents

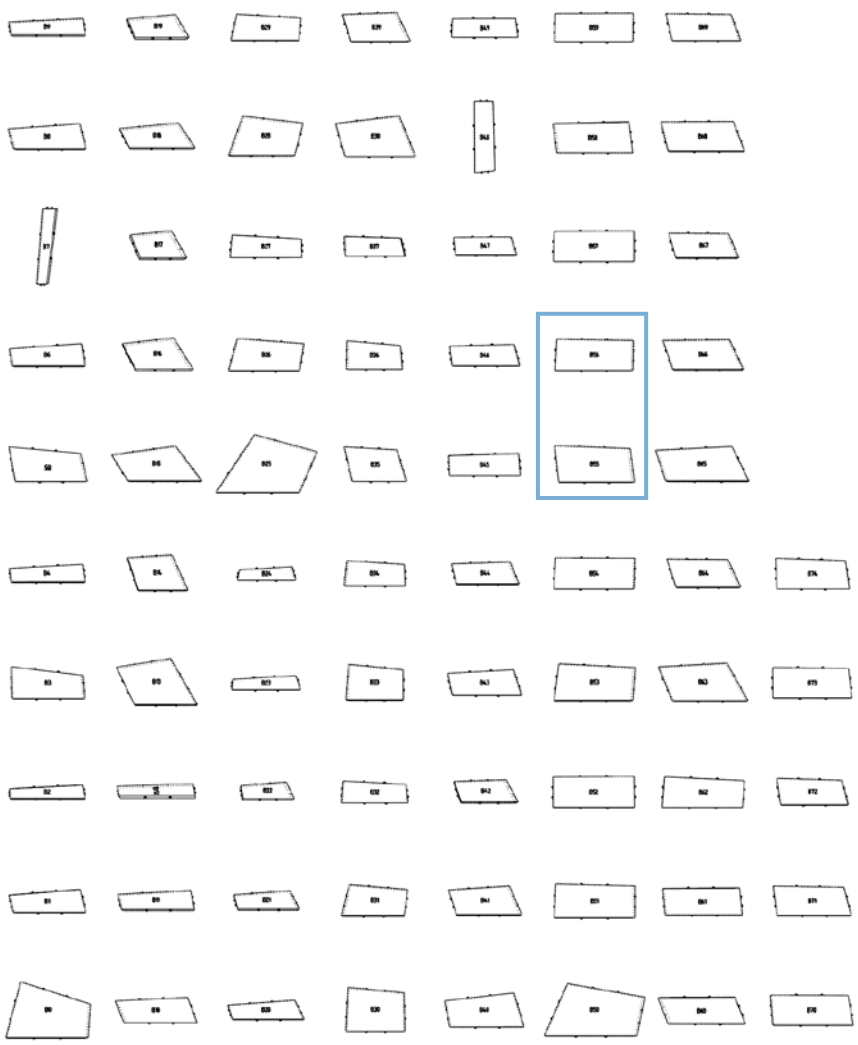
Compression-only structures rely entirely on geometry, gravity, and friction to eliminate tensile stress. This is the exact mechanical principle behind ancient Roman masonry arches and traditional Inuit igloos. This allows discrete blocks to span spaces without mortar or reinforcement.

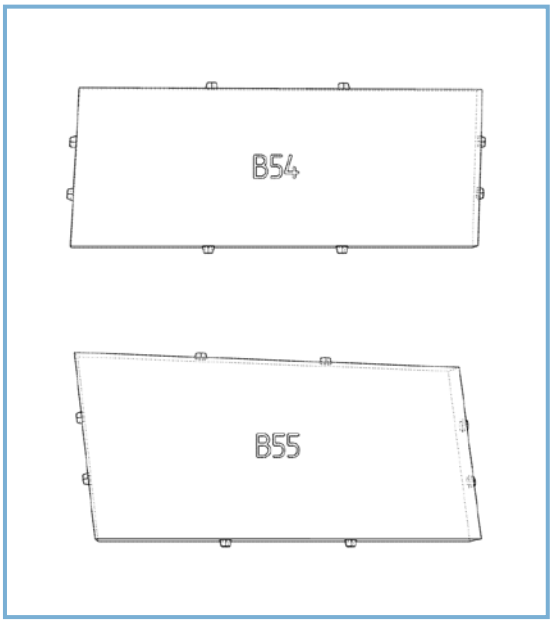
Some of the Contemporary Precedents:

Armadillo Vault: Spanned 16 meters using 399 unreinforced, dry-jointed limestone blocks, optimized through algorithmic form-finding.

Bricktopia: Merged parametric funicular analysis with traditional Catalan vaulting to create constructible, unreinforced masonry shells.







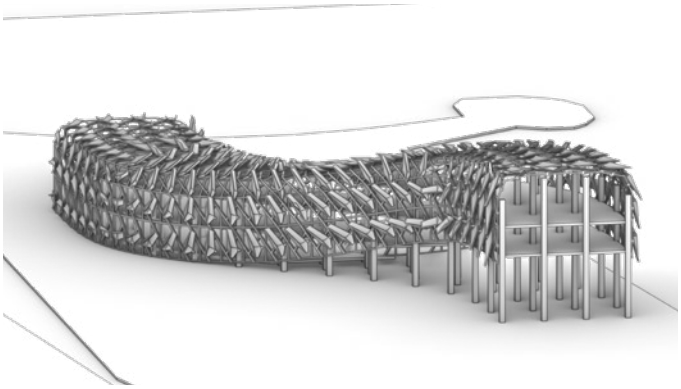
To transition to physical assembly, the workflow automates the tracking of hundreds of unique blocks. Using UV center-points, the script procedurally generates text profiles (e.g., A-01 for female blocks, B-01 for male). These are extruded and subtracted, engraving alphanumeric identifiers into every top face.

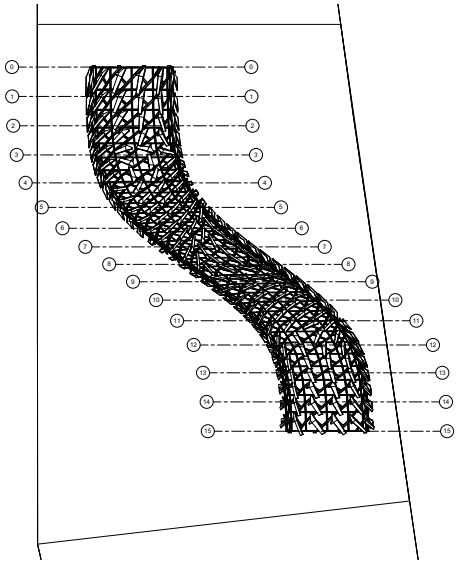
Because digital fabrication requires flat orientation, the script extracts each block's planar bottom surface and calculates a transformation matrix. Using the Orient component, 3D geometries are arrayed onto the XY plane. This directly outputs pre-numbered, watertight STLs, bypassing manual file preparation.

Dragon Scale Pavilion

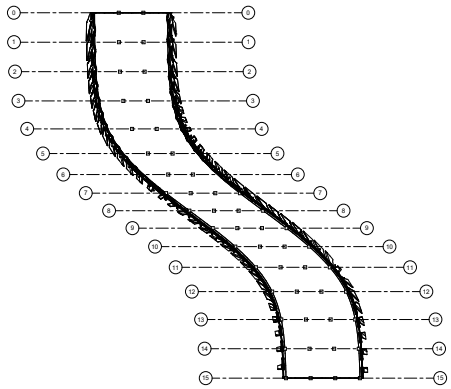
Rhino Inside Revit Workflow

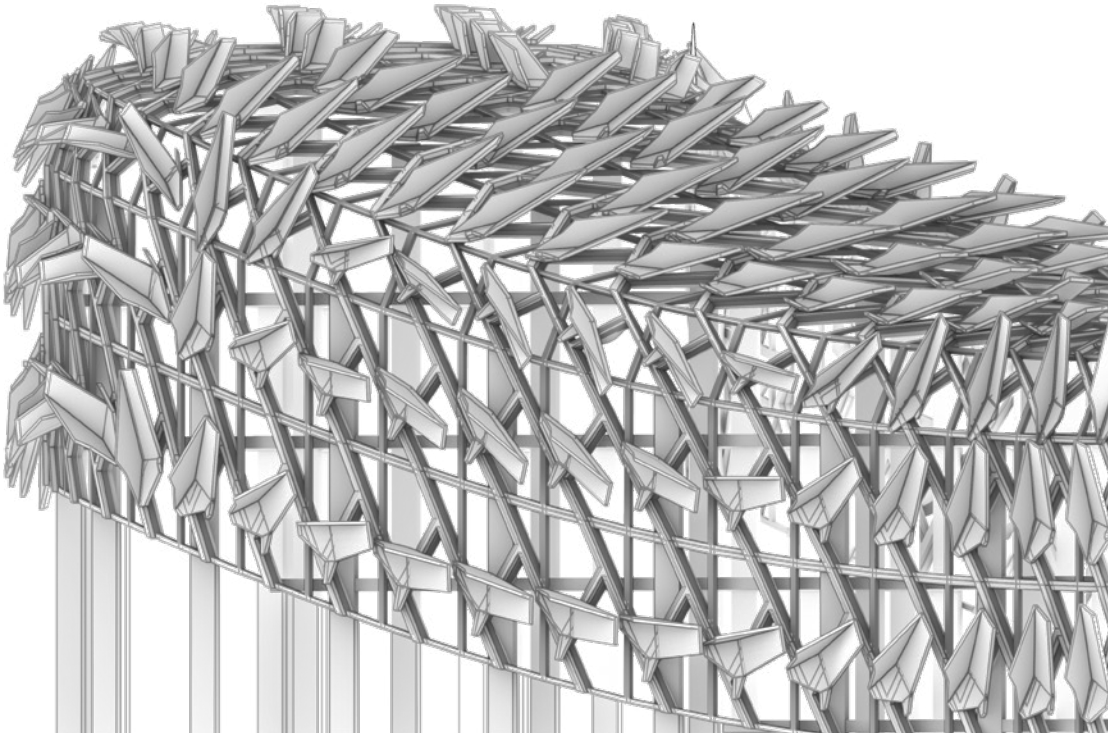
This project explores the direct integration of algorithmic modeling with Building Information Modeling (BIM). Utilizing Rhino.Inside.Revit, the workflow bypasses traditional file-export limitations, running Grasshopper directly within the Revit environment. This enables the automated generation of native BIM elements driven by complex parametric logic, maintaining dynamic, bi-directional data flow from early massing to detailed documentation.





The computational pipeline begins by extracting spatial coordinates to automatically generate native Revit structural grids, levels, and columns. A parametric algorithm then evaluates the building envelope, deploying adaptive curtain walls and complex facade systems directly onto the Revit host geometry. Finally, terrain data is processed to construct accurate landscape topographies. This workflow ensures all algorithmic geometry translates instantly into schedulable, fully parameterized BIM data.





Contacts

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saraelimam.wordpress.com

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